Signal Processing Methods for Lubrication Condition Monitoring of Ball Bearings: A Systematic Review

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Abstract

Write abstract last, after all sections are complete.

Contents

1 Introduction

Rolling element bearings are fundamental components of virtually all rotating machinery, from electric motors and pumps to wind turbines, machine tools, and rail vehicles. Their reliable operation underpins industrial productivity across manufacturing, energy, and transport sectors. Lubrication is the critical enabler of bearing function: a thin film of oil or grease separating the rolling elements from the raceways reduces contact friction, carries heat away from the loaded zone, and prevents the metal-to-metal contact that initiates wear and fatigue. When lubrication fails – whether from starvation, viscosity degradation, or contamination – asperity contact increases, heat generation intensifies, and wear proceeds at a rate that can reduce bearing service life from tens of thousands of hours to a small fraction of its design value. Improper lubrication is estimated to cause between 30% and 80% of rolling element bearing failures [1–4], consistently ranking among the leading causes of premature failure ahead of mechanical overload and installation errors [5].

The economic consequences of lubrication-related bearing failure extend well beyond the replacement cost of the component. Unplanned failure of a bearing in a continuous process plant – paper mill, steel plant, petrochemical facility – can halt an entire production line, with downtime costs that are typically an order of magnitude greater than the cost of the bearing itself. In wind energy, main shaft and gearbox bearing failures are among the most costly maintenance events; at offshore installations, crane mobilisation and vessel hire costs can dwarf the hardware expenditure, and weather-limited access windows mean that a single unplanned replacement can represent multiple days of lost generation and tens of thousands of euros in intervention costs [6]. Across industry, lubrication has been identified as a primary driver of rotating machinery maintenance spend, and the reduction of premature failures attributable to inadequate lubrication represents one of the largest available levers for improving asset availability and reducing lifecycle maintenance costs.

Industrial relubrication practice, however, remains predominantly time-based: lubricant is applied at fixed calendar or operating-hour intervals regardless of the actual condition of the lubricant in the bearing. This approach is systematically suboptimal in both directions. Over-lubrication – replenishing when existing lubricant remains adequate – causes churning losses, elevated operating temperature, seal pressurisation, and in grease-lubricated bearings can disrupt the channelling behaviour that grease develops under normal operation, itself accelerating degradation. Under-lubrication – where the fixed interval is too long for actual operating conditions – leads to film thinning, progressive asperity contact, and accelerated wear that is invisible to operators until failure symptoms emerge [6]. The correct relubrication interval and quantity are functions of bearing geometry, rotational speed, load, ambient temperature, and lubricant type and state – a combination of variables that no fixed schedule can optimally accommodate, and that varies continuously with machine operating conditions. Surveys consistently report that both over- and under-lubrication are prevalent in industrial practice, each accounting for a meaningful share of lubrication-related failures [6].

Condition-based relubrication (CBR) – lubricating in response to a measured indication of lubricant condition rather than a fixed schedule – eliminates both failure modes simultaneously. CBR requires continuous, non-disruptive online monitoring of whether the lubricant film in the bearing is adequate for current operating conditions. This is the core problem of lubrication condition monitoring (LCM). LCM encompasses the estimation of lubricant film thickness, viscosity adequacy, degradation state (chemical and rheological aging), and contamination by wear debris or environmental ingress – the quantities that collectively determine the protective capacity of the lubricant. Accurate online LCM would replace time-based schedules with maintenance decisions grounded in the actual lubricant state, enabling operators to lubricate at the right time in the right quantity [6].

Despite its practical importance, LCM has received substantially less research attention than vibration-based bearing fault detection and remaining useful life (RUL) estimation, which have been studied for decades and are now widely deployed in industry. The barrier to wider LCM deployment is not primarily a sensor hardware limitation: sensing modalities capable of detecting lubrication-related signals – acoustic emission, passive ultrasound, vibration, electrical capacitance, active ultrasonic reflectometry – have been demonstrated across a substantial body of laboratory research. The primary barrier is signal processing: the methods for extracting reliable, operating-condition-invariant lubrication condition estimates from sensor data are immature, fragmented across sensing modalities, and rarely validated beyond controlled laboratory conditions. This review addresses that gap directly.

Online monitoring, as used throughout this review, refers to continuous data acquisition from a bearing in operation without interruption or lubricant extraction. *In-line* monitoring uses sensors integrated into the lubricant circulation system and likewise requires no bearing stoppage. Both modes are within the scope of this review. *At-line* monitoring (near-process analysis requiring brief interruption) and *offline* monitoring (laboratory analysis of extracted samples) are excluded.

1.1 Monitoring Flow

An LCM program with clear sequential steps is defined in [6]. This, however, considers actual sampling and direct analysis of the lubrication. We define the following steps for online LCM which relies on information gathering during bearing operation:

1. **Sensing**: observation and data collection from a bearing in operation through sensors.
2. **Signal processing**: signal processing and machine learning, which prepares the data for the downstream task of lubrication condition assessment. This step might contain dimensionality reduction through feature extraction or it might preserve the time-series structure of the data.
3. **Lubrication condition assessment**: classification or regression on the processed data to infer the current lubrication condition in terms of lubrication regime, film thickness, degradation state or contamination; elaborated in Section 3.
4. **Maintenance decision**: a concrete relubrication strategy based on the lubrication condition assessment – when and how much to lubricate to maintain or restore adequate lubrication level.

The steps in the LCM-based maintenance pipeline are visualised in Figure 1. LCM might be concerned with all steps 1–4 or just steps 1–3 with step 4 as a downstream task. Steps 1–3 – which include signal processing and lubrication condition assessment sitting between the raw sensor data and the maintenance decision – are the core focus of this review.

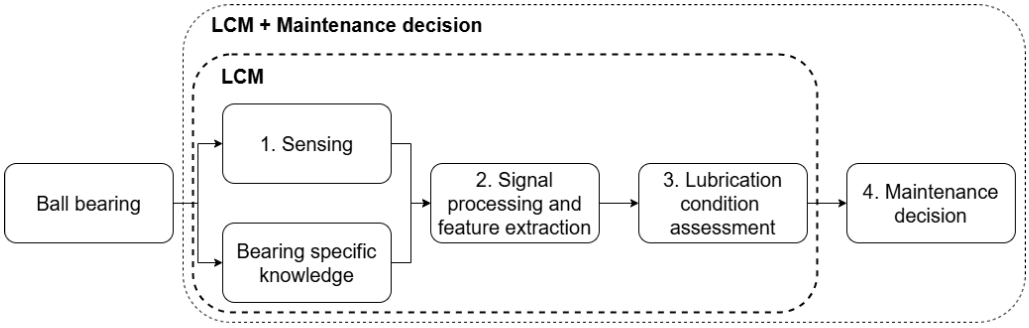

Figure 1: Steps in online LCM of a running ball bearing. Signal processing sits between the raw sensor data and the lubrication condition assessment.

1.2 Existing Reviews and Their Limitations

A number of reviews address related topics in LCM. Wakiru et al. [6] review LCM for maintenance decision support, covering direct lubricant analysis by invasive methods. Shang et al. [7] review film thickness estimation by optical, electrical, and ultrasonic methods (active and/or invasive). Wang et al. [8] review bearing condition monitoring including lubricant film thickness and wear debris detection, covering primarily active or invasive methods while mentioning passive ultrasound and acoustic emission measurements for condition monitoring. Hansen et al. [9] review calibration for ultrasonic reflectometry (active). Cen et al. [10] and Becker-Dombrowsky et al. [11] review electrical capacitance and impedance methods (active). Dai et al. [2] review in-situ film thickness measurement by capacitance (active). Zhu et al. [12, 13] review online lubricant sensors and oil monitoring methods (active and invasive). Sun et al. [14] review debris monitoring methods (active). Dou et al. [15] review ultrasonic reflectometry for film thickness (active).

Three limitations are shared by all existing reviews. First, none provides a systematic taxonomy of signal processing methods across sensing modalities; signal processing is treated as a secondary detail beneath the sensing method taxonomy. Second, existing reviews overwhelmingly focus on active or invasive methods, leaving passive non-invasive methods – which offer the greatest practical advantages for industrial deployment – underrepresented. Third, no review identifies signal processing as the primary bottleneck limiting the advancement of LCM from controlled laboratory demonstrations to industrial deployment.

1.3 Contribution and Scope of This Review

This review provides the following contributions:

1. **A systematic taxonomy of signal processing methods for LCM**, organised by SP method family and applied across all sensing modalities – the first such taxonomy in the LCM literature.

2. **A comprehensive comparative analysis** of sensing and SP method combinations across four dimensions: lubrication fault type sensitivity, operating condition sensitivity, deployment requirements, and technology maturity.

3. **A structured identification of SP-centric research gaps** that limit the advancement of LCM to industrial practice, with specific future research directions for each gap.

4. **Coverage of all sensing modalities** – active and passive, invasive and non-invasive – enabling a complete picture of the field, with greatest depth given to the passive non-invasive methods that are most relevant to industrial retrofit applications.

This review covers the literature on online and in-line LCM of rolling element bearings (primarily ball bearings), with a focus on studies that describe signal processing methods in sufficient detail to assess their assumptions, limitations, and applicability. Studies concerned exclusively with bearing fault detection without a lubrication-specific focus, and studies on journal bearings or non-bearing tribological contacts, are excluded. The search covers literature up to early 2026.

1.4 Paper Organisation

Section 2 describes the systematic literature search process and inclusion criteria. Section 3 reviews the theoretical background of lubrication physics, degradation mechanisms, and models. Section 4 discusses the physical origins of sensor signals in the context of LCM. Section 5 provides a taxonomy and description of sensing methods, including the formal definitions of the active/passive and invasive/non-invasive classifications used throughout this review. Section 6 is the primary contribution: a systematic review of signal processing methods organised by SP family. Sections 3–6 together cover the complete pipeline from physical signal origins to signal processing methods. Section 7 presents the comparative analysis. Section 8 identifies research gaps and future directions. Section 9 concludes.

2 Systematic Literature Search

2.1 Search Strategy

The literature search was conducted in January–February 2026 using the databases Google Scholar, ScienceDirect, Scopus, and ResearchGate, supplemented by citation graph exploration via Connected Papers to identify relevant works not captured by keyword search alone. Boolean search strings were constructed from combinations of the following keyword groups:

- *Application domain:* lubrication, lubrication condition monitoring, film thickness, lubrication degradation, lubricant starvation, grease condition, oil condition.
- *Component:* ball bearing, rolling element bearing, roller bearing.
- *Sensing modality:* vibration, acoustic emission, ultrasound, sound, microphone, thermal, infrared, electrical, capacitance, impedance, optical, magnetic, inductive.
- *Signal processing:* signal processing, feature extraction, time domain, frequency domain, wavelet, machine learning, deep learning, neural network, regression, classification.
- *Monitoring mode:* online, in-line, in-situ, condition monitoring.

2.2 Inclusion and Exclusion Criteria

Inclusion criteria	**Exclusion criteria**
Study concerns lubrication condition in rolling element bearings (including ball bearings, cylindrical roller bearings, tapered roller bearings, or slew bearings)	Study concerns journal bearings, thrust washers, or non-bearing lubricated contacts exclusively
Sensing method and signal processing approach are described in sufficient detail to assess the physical principle, processing operations, and output	Study provides only qualitative descriptions of sensing and processing without sufficient methodological detail
Monitoring is online, in-line, or in-situ (data collected during bearing operation)	Monitoring is exclusively offline (lubricant laboratory analysis without online data collection)
Study focuses on lubrication condition as the primary monitoring target (film thickness, viscosity, degradation, contamination, or regime)	Study concerns bearing fault detection or RUL estimation with no specific lubrication condition focus
English-language peer-reviewed journal papers, conference proceedings or manuals from industrial applications for LCM	Non-peer-reviewed sources (technical reports, white papers, patents)

Table 1: Inclusion and exclusion criteria for the systematic literature search.

2.3 Search Results and PRISMA Flow

A total of 131 sources were initially identified through the literature search. After removal of duplicates across databases, 48 sources were identified as treating the topic of online or in-line LCM for rolling element bearings. Of these, 33 treated monitoring with active or invasive sensing methods and 20 treated passive and non-invasive sensing methods (some sources address both). All 48 are included as the primary basis for this review; the broader set of 131 sources provides supporting context for theoretical background, signal processing methodology, and industrial applications.

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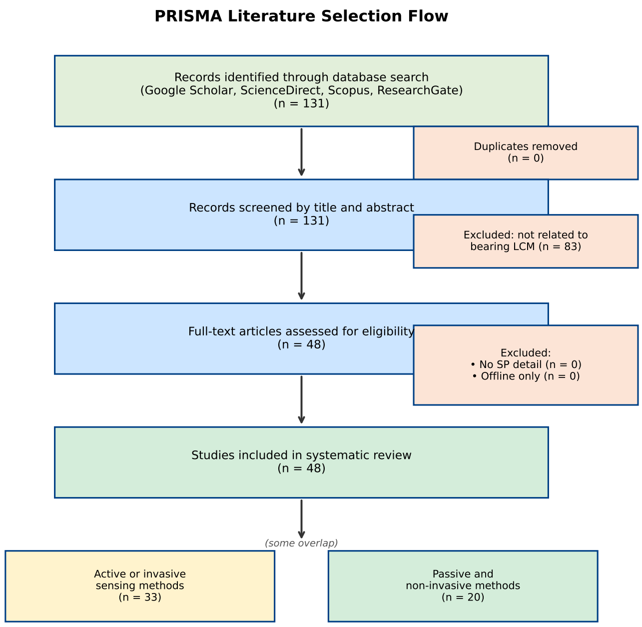

Figure 2: PRISMA-style literature selection flow diagram. Numbers reflect the search conducted in January–February 2026.

3 Lubrication Physics, Models and Degradation Mechanisms

Multiple theoretical models of lubrication exist which seek to estimate the lubrication film thickness in running ball bearings and the sufficiency thereof under specific operating conditions and with different lubricants. However, a complete understanding and model of the physico-chemical properties of lubricants and the lubricant film formation in ball bearings has yet to be formulated [10] [2] [16].

The following subsections present the theoretical models most commonly used in LCM research and the degradation mechanisms that affect lubrication conditions. The theoretical models serve two distinct roles: they provide the physics-based reference against which sensor measurements are calibrated, and they establish which physical quantities – film thickness, friction, heat generation, and elastic wave emission – can in principle be observed by each class of sensing method. This connection between lubricant physics and observable signal is the foundation for understanding which signal processing operations are meaningful for each modality. The degradation mechanisms are important for understanding the physical origins of signals and the specific monitoring targets that signal processing methods should be optimised for.

3.1 Lubrication Regimes

Lubrication films formed in ball bearings under operation are generally classified into three regimes depending on the physical contact between balls and raceways and how the bearing load is distributed:

- **Boundary lubrication**: the entire load is carried by surface-to-surface contact (ball to raceway). This regime produces the highest friction, arising solely from sliding of asperities against each other. Wear rate is high, and heat generation and elastic stress-wave emission are correspondingly intense.
- **Mixed lubrication**: load is shared between a thin lubricant film and surface asperity contacts. Friction and wear are reduced compared to the boundary regime but asperity contacts continue to generate transient stress waves detectable by acoustic and vibration sensors.
- **Elastohydrodynamic lubrication (EHL)**: all load is carried by a lubricant film between the balls and raceway. The high pressure building up in the contact zone causes elastic deformation of the solid materials, sustaining the film. Friction is at its minimum for full-film EHL and rises again at excessive viscosities due to viscous shear losses. Wear is minimal and elastic wave emission is negligible, making this the ideal operating regime for bearing longevity.

The progression through these regimes as lubrication conditions deteriorate determines both the nature and the intensity of the signals accessible to passive sensing methods.

3.2 The Stribeck Curve and Its Significance for Signal Processing

The Stribeck curve describes the relationship between the friction coefficient and the Hersey number $\eta n/p$, where η is the dynamic viscosity, n is the rotational speed, and p is the contact pressure. As the Hersey number decreases – due to lower speed, lower viscosity, or higher load – the bearing transitions from the EHL regime, through mixed lubrication, into boundary lubrication, and the friction coefficient rises sharply. The curve is illustrated in Figure 3

The Stribeck curve is directly relevant to signal processing because the friction coefficient governs the amplitude of signals detectable by passive methods: higher friction produces stronger elastic wave emission (detectable by acoustic emission and ultrasound sensors), higher heat generation (detectable by temperature and infrared sensors), and increased mechanical vibration (detectable by accelerometers and microphones). A bearing operating in the EHL regime near the Stribeck minimum will produce weaker passive signals than one in boundary lubrication. This has two important signal processing consequences: first, monitoring under optimal lubrication (and in a neighbourhood of the optimum) is inherently a low-SNR problem; second, any feature that is strongly correlated to friction can in principle serve as a lubrication condition indicator. Hou and Zhang [17] demonstrate this explicitly by mapping acoustic emission amplitude and energy onto lubrication regime clusters, with the joint distribution in a two-dimensional (amplitude, energy) space discriminating boundary, mixed, and EHL conditions.

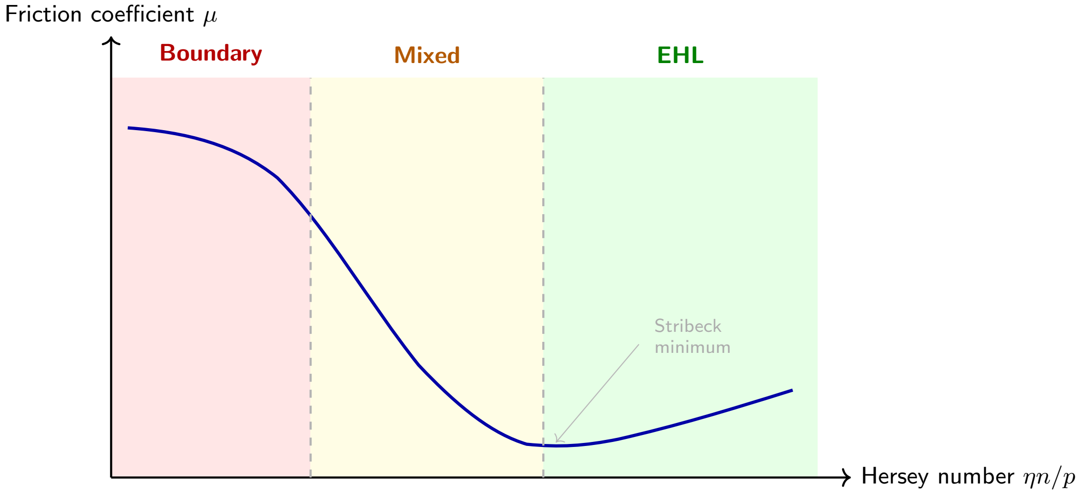

Figure 3: Schematic Stribeck curve showing friction coefficient μ as a function of Hersey number $\eta n/p$, where η is dynamic viscosity, n rotational speed, and p contact pressure. As the Hersey number decreases – due to lower speed, lower viscosity, or higher load – the bearing transitions from full-film EHL through mixed lubrication into boundary lubrication, with a sharp rise in friction coefficient. The Stribeck minimum marks the onset of full-film lubrication.

3.3 Hamrock–Dowson Model

The Hamrock–Dowson model is the most widely used empirical model for calculating the minimum film thickness, h_m, in ball bearings. It is given by

$$\frac{h_m}{R_x'} = 3.63\, U^{0.68}\, G^{0.49}\, W^{-0.073}\, (1 - e^{-0.68k}), \tag{1}$$

where

- R_x' and R_y' are the effective radii in the direction parallel and transverse to the lubricant entrainment, respectively,
- $U = \eta_0 u_e/(E' R_x')$ is a dimensionless speed parameter, with u_e the lubricant entrainment speed and E' the effective elastic modulus,
- $W = w_z/(E' R_x'^{\,2})$ is a dimensionless load parameter, with w_z the applied normal load,
- $G = \alpha^* E'$ is a dimensionless material parameter, with α^* the pressure-viscosity coefficient, and
- $k = 1.03(R_y'/R_x')^{2/\pi}$ is the elliptic parameter.

Further derivation and physical motivation of these parameters can be found in [10, 18–20]. In the LCM context, the Hamrock–Dowson model is most often used not to predict film thickness directly, but to generate reference values of h_m against which sensor-derived estimates can be calibrated [1, 17, 21].

3.4 Λ-Ratio

The Λ ratio is the ratio between the lubrication film thickness and the composite surface roughness of the bearing components, and is widely used to assess the lubrication regime. It is defined as

$$\Lambda = \frac{h_m}{\sqrt{R_{q1}^2 + R_{q2}^2}}, \tag{2}$$

where h_m is the minimum film thickness and R_{q1}, R_{q2} are the root-mean-square roughness values of the two contacting surfaces. Lubrication regimes are conventionally assigned as: $\Lambda < 1$ (boundary), $1 \leq \Lambda \leq 3$ (mixed), $\Lambda > 3$ (EHL). The Λ ratio is the primary physical target variable in some LCM studies that use sensor signals to estimate the lubrication regime [1, 17, 22].

Hansen et al. [18] note that the conventional Λ ratio has shortcomings in the EHL regime arising from elastic asperity deformation, and propose an improved Λ^* ratio:

$$\Lambda^* = \frac{h^*}{z_0} = \frac{h_m + \delta_a}{z_0}, \tag{3}$$

where δ_a is the elastic asperity deformation in EHL and z_0 is the relaxed-state height of the surface profile.

3.5 κ-Ratio

The κ ratio is a model-based indicator of lubricant viscosity adequacy for a specific bearing under specific operating conditions. It is defined as

$$\kappa = \frac{\nu}{\nu_1}, \tag{4}$$

where ν is the kinematic viscosity of the applied lubricant under the operating conditions and ν_1 is the reference kinematic viscosity calculated from the ISO 281 standard for the bearing diameter and rotational speed [1]. Interpretation guidelines are :

Find citation for κ thresholds (ISO 281 or SKF/FAG catalogue)

- $\kappa \leq 0.1$: insufficient lubrication; load carried almost entirely by asperity contact.
- $0.1 < \kappa \leq 1$: insufficient lubrication; metallic surface-to-surface contact present.
- $1 < \kappa \leq 4$: adequate lubrication.
- $4 < \kappa$: excessive viscosity; higher viscous friction losses.

Given the fact that κ can be tied to lubrication conditions it can also be tied to the Stribeck curve and lubrication regimes. Jakobsen et al. [1, 23] use κ as the regression target for data-driven models trained on vibration and ultrasound features, linking directly to the ISO standard and enabling quantitative LCM without invasive lubricant sampling. It should be noted that κ is time-invariant for a given bearing and operating condition; it therefore captures viscosity inadequacy if the conditions remain constant but does not account for lubricant degradation or contamination that develops over time.

3.6 Lubricant Degradation Mechanisms

For oil-lubricated bearings, degradation proceeds primarily through: thermal and oxidative aging (which increases viscosity temporarily, then reduces it as polymer chains break down), contamination by wear particles or external debris, and oil starvation from insufficient supply or replenishment. For grease-lubricated bearings the degradation picture is more complex. Grease consists of a base oil held in a thickener matrix; degradation involves base-oil bleed-out (reducing available lubricant), mechanical working of the thickener (reducing consistency and channelling behaviour), thermal degradation (affecting both base oil and thickener), and contamination [24–26]. Cen et al. [27] specifically study the effect of thermal aging on grease film thickness in ball bearings and show measurable film thickness reduction with aging time. These differing degradation mechanisms mean that signal processing pipelines tuned to detect oil starvation may not transfer directly to grease monitoring applications and vice versa, a gap explicitly identified in Section 8.

4 Physical Origins of Measurable Signals

The connection between lubricant physics and measurable signals is the key link between the theoretical models in Section 3 and the sensing and signal processing methods reviewed in Sections 5 and 6. Table 2 maps each physical mechanism through which lubrication condition becomes observable to the sensing modality that captures it, the frequency band available for signal processing, the LCM quantities it encodes, and the principal practical constraint on its use.

Physical mechanism	Signal type	Sensing modality	Freq. range at sensor	LCM quantities encoded	Principal constraint
Asperity micro-contact and micro-fracture, and lubricant-traction-driven sliding friction, at ball–raceway interface under insufficient lubricant film	Transient elastic stress waves	AE sensors; passive ultrasound (§5.2.1, §5.2.3)	20–500 kHz (path-limited) [28, 29]	Lubrication regime (Λ); starvation severity; abrasive contamination events [17, 22, 30, 31]	Path attenuation 30–60 dB at 500 kHz through housing; usable bandwidth geometry- and mounting-specific [28, 29]
Rolling/sliding friction and impulsive excitation from asperity contacts at ball–raceway and ball–cage interfaces	Broadband structural vibration; radiated audible sound	Accelerometers; microphones (§5.2.2, §5.2.4)	Vibration: 0–20 kHz; Sound: 20 Hz–20 kHz	Lubrication adequacy (κ, Λ); starvation onset [3, 23, 32]	Standard industrial accelerometers limited to <10 kHz; LCM-informative content predominantly >30 kHz [1]
Viscous shear heating and boundary-lubrication friction energy dissipation	Bearing temperature rise	Thermocouples; IR thermography (§5.2.5)	DC–10 Hz (thermally diffusion-limited)	Sustained lubrication inadequacy; long-term friction regime shift [33, 34]	Thermal response lags lubrication onset by seconds to minutes; insensitive to transient or intermittent starvation events
Lubricant film thickness and dielectric state alter electrical properties of bearing contact	Change in capacitance, impedance, or resistance	Electrical sensors (§5.3.2)	DC–10 MHz (excitation dependent)	Film thickness (h_m, Λ); lubricant degradation state; water ingress [10, 11, 35]	Bearing electrical isolation and shaft brush or slip-ring contacts required; not retrofittable to standard machines
Partial reflection of ultrasonic pulses at the lubricant–raceway interface (acoustic impedance mismatch)	Reflected ultrasonic pulse; reflection coefficient	Active ultrasonic reflectometry; SAW sensors (§5.3.1, §5.3.3)	1–100 MHz (applied excitation)	Film thickness (h_m) via spring-layer model; lubricant bulk modulus [36–38]	Double-path transmission loss; precise coupling, alignment, and reference calibration required; not retrofittable [9]
Magnetic field perturbation by ferromagnetic or conductive wear particles in lubricant flow	Inductance change; eddy current signature	Inductive/magnetic debris sensors (§5.3.5)	DC–kHz (debris transit)	Ferrous wear debris count and size; contamination severity [39]	Applicable only to oil-circulating systems; inapplicable to grease-lubricated or sealed bearings
Optical interaction of light with lubricant film (interference, fluorescence) or suspended debris (scattering)	Interference fringe pattern; scattered or attenuated light intensity	Optical interferometry; LIF; machine-vision (§5.3.4)	Optical (THz–PHz)	Film thickness (h_m) via interferometry; lubricant composition via LIF; debris count and size via machine vision [7, 40, 41]	Interferometry and LIF require transparent contact zone (purpose-built test rigs only); machine vision requires in-line optical access to lubricant flow

Table 2: Physical mechanisms linking lubrication conditions in ball bearings to measurable signals. 'Freq. range at sensor' is the band available to signal processing methods after transmission through the structural path from the contact zone to the sensor; see Section 5.1. 'LCM quantities encoded' identifies the lubrication-condition-relevant information carried by each signal, directly connecting physical origin to the sensing discussion in Section 5 and the comparative analysis in Section 7. Sensing modality cross-references refer to subsections of Section 5.

Recent mechanistic work refines the asperity-contact picture in the first row of Table 2. Zhang et al. [31] couple a roller-bearing dynamics model with an AE generation model and show, validated on greases of distinct viscosities, that AE under mixed lubrication arises from two superposed sources rather than asperity contact alone: AE driven by the normal load on asperities scales with the square root of rotational speed, while AE driven by lubricant-traction sliding friction scales quadratically with speed. Which mechanism dominates is lubricant-dependent – high-viscosity grease, with its higher traction coefficient, produces friction-dominated, quadratically increasing AE, whereas low-viscosity grease remains asperity-load-dominated. This provides a mechanistic explanation for why the AE-lubrication relationship is configuration- and lubricant-specific, as observed empirically in Section 5.2.1.

The physical mechanisms in Table 2 directly motivate the sensing method taxonomy in Section 5, the signal processing choices reviewed in Section 6, and the comparative fault-type sensitivity analysis in Table 4.

5 Sensing Methods

Sensing methods for online LCM of ball bearings are classified along two independent axes. A method is **active** if it supplies external energy into the bearing system to probe its response; it is **passive** if it relies solely on signals generated by the bearing under operation. A method is **invasive** if it requires unobstructed access to the lubricant film, direct physical contact with the lubricant, sensors mounted inside the bearing, or structural modifications to the bearing or its housing; it is **non-invasive** if none of these requirements apply. Online and in-line monitoring modes are defined in Section 1.

These two axes define four categories illustrated in Figure 4: active/invasive (e.g., optical interferometry, ultrasonic reflectometry), active/non-invasive (e.g., external ultrasonic pulse-echo with external transducer), passive/invasive (e.g., direct lubricant sampling), and passive/non-invasive (e.g., acoustic emission, vibration, thermal monitoring). The passive/non-invasive category is of particular industrial relevance because its methods can be applied to bearings in service without modification and without interrupting operation. The present review covers all four categories for completeness but gives greatest depth to the passive/non-invasive methods.

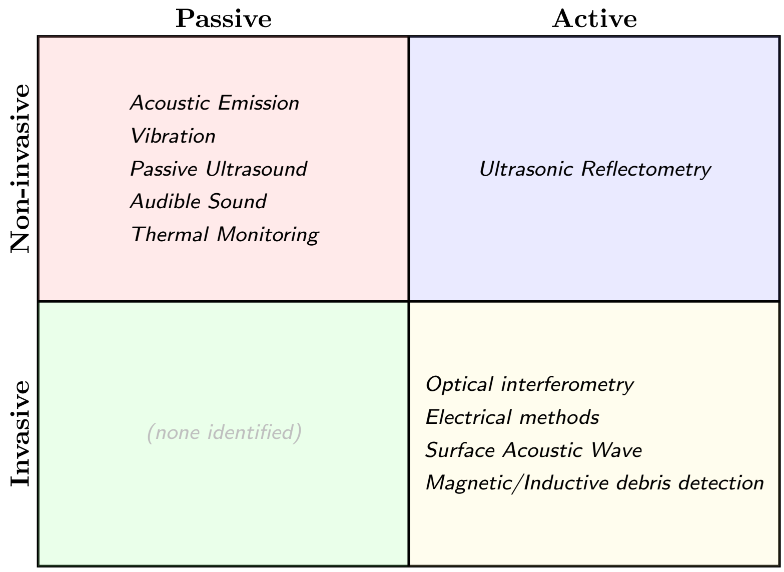

Figure 4: Classification of sensing methods for lubrication condition monitoring along two independent axes: active vs. passive signal acquisition, and invasive vs. non-invasive installation requirements. The passive/invasive quadrant contains no identified methods: sensing methods that require physical access to the lubricant (invasive) almost invariably do so to supply or measure a controlled excitation (active).

The subsections below describe each sensing modality with a standardised structure: (i) physical principle, (ii) what lubrication information it is sensitive to, (iii) sensor hardware and technology considerations, and (iv) installation and operational characteristics. Signal processing methods are treated in depth in Section 6. A factor common to all modalities is the transmission path between the signal origin and the sensor, which is treated first as it governs which signals are accessible at the sensor regardless of modality.

5.1 Signal Transmission Path and Frequency-Dependent Attenuation

All online sensing methods for LCM share a fundamental constraint: the sensor cannot be placed at the signal origin. The bearing contact zone – where asperity contacts, lubricant film formation, and wear events occur – is physically inaccessible to a sensor. Any measurable signal must first propagate from the contact zone to the sensor through a series of solid media and mechanical interfaces. Understanding this transmission path is not merely a practical installation concern; it directly constrains which signal processing operations are valid, which frequency bands are informative, and how results from laboratory rigs transfer to industrial deployments.

For a sensor mounted on the bearing housing, the signal path from the ball–raceway contact zone crosses four sequential segments: the bearing element material, the outer ring–housing bore press-fit interface, the housing wall, and the sensor coupling layer. Each material and interface boundary introduces frequency-dependent attenuation; cumulative losses over a typical path through the housing material and press-fit interface mean that a 500 kHz AE signal may arrive at the sensor attenuated by 30–60 dB relative to the source [28, 29] – a range that spans the difference between a detectable signal and the noise floor. For active methods such as ultrasonic reflectometry and SAW sensing, the path is traversed twice and losses are compounded. The transmission path therefore acts as a *frequency-dependent low-pass filter* whose effective bandwidth decreases as path length increases or more interfaces are crossed. This path is illustrated in Figure 5.

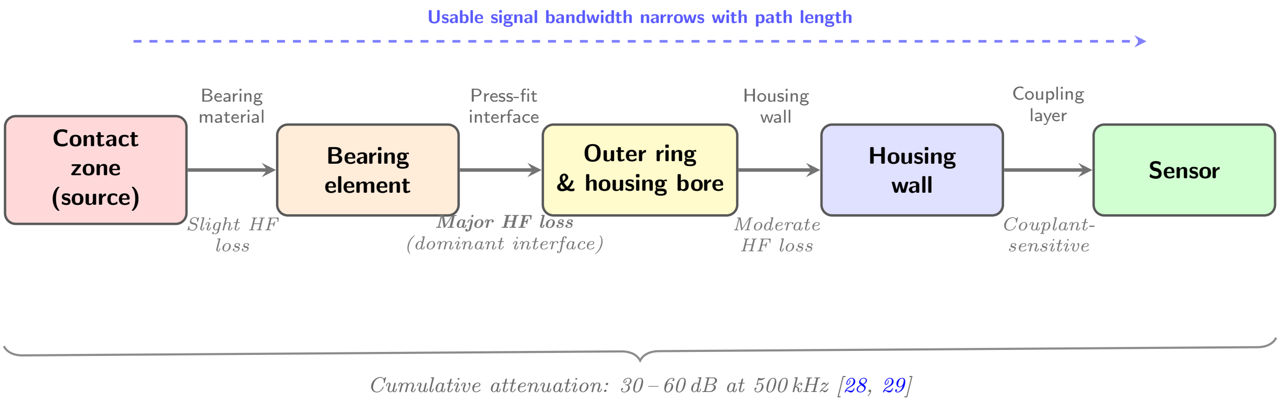

Figure 5: Signal transmission path from the ball–raceway contact zone to a housing-mounted sensor. The press-fit interface between the outer ring and housing bore is the dominant attenuation element for high-frequency content. Cumulative path losses of 30–60 dB at 500 kHz render the usable frequency bandwidth at the sensor geometry-, mounting-, and material-specific, limiting the transferability of signal processing configurations across experimental setups.

This has two direct consequences for signal processing.

First, *frequency-domain configurations derived from one experimental setup do not automatically transfer to another*. The optimal bandpass for spectral kurtosis, the resonant frequencies of the housing structure, and the AE frequency content observable at the sensor are all geometry-specific. This limits the portability of SP configurations tuned on laboratory test rigs to industrial deployments with different sensor placements, housing geometries, or coupling conditions.

Second, *sensor placement is a signal processing decision, not only an installation practicality.* Mounting a sensor across a bolted flange, a material transition, or a long structural path can render high-frequency AE content undetectable regardless of sensor sensitivity. Where possible, sensors should be mounted on a flat, clean surface with a direct structural path to the bearing bore, minimising the number of interfaces and the path length. For non-contact microphone monitoring, the additional transmission from the housing surface into air is highly frequency-dependent: audible sound (20 Hz–20 kHz) propagates relatively efficiently, but ultrasonic frequencies (>40 kHz) undergo rapid geometric spreading and air absorption [42], limiting non-contact acoustic monitoring to the audible and near-ultrasonic range.

5.2 Passive and Non-Invasive Methods

5.2.1 Acoustic Emission

Acoustic emission (AE) refers to transient, high-frequency elastic stress waves – typically in the range 20 kHz to 1 MHz – spontaneously generated within the bearing material and contact interfaces as a result of tribological processes. These processes include asperity contact and micro-fracture under inadequate lubrication, crack initiation and propagation under cyclic contact stress, wear particle generation, and frictional energy release at rolling and sliding contacts [17, 43, 44]. AE sensors are piezoelectric transducers mounted on the bearing housing or an adjacent structural component and capture these stress waves non-invasively.

AE is sensitive to the lubrication regime: as the specific film thickness Λ decreases from EHL toward boundary lubrication, asperity contacts increase in frequency and severity, generating more AE energy. Cornel et al. [22] demonstrate a direct inverse correlation between AE amplitude and Λ in cylindrical roller bearings under controlled lubrication conditions. Hou and Zhang [17] show that statistical amplitude and energy features from AE signals in thrust ball bearings map onto distinct lubrication regime clusters and can be used to estimate Λ with 89.3% accuracy (see Section 6.1.1 for the feature definitions). Miettinen et al. [30] show that the pulse count rate increases monotonically with grease starvation and differentiates among ten grease compositions. An important exception to the monotonic amplitude-lubrication relationship is the non-monotonic energy response observed by Hou and Zhang, in which transitioning from dry contact to low-viscosity lubrication initially *decreases* AE energy before it rises again with increasing viscosity due to viscous damping forces on the bearing ring. This highlights that AE-lubrication relationships are configuration-specific and require calibration. Zhang et al. [31] provide a mechanistic account of this lubricant-dependence: in cylindrical roller bearing tests with greases of distinct viscosities, they show AE arises from two superposed sources – normal-load asperity contact (RMS scaling with $\sqrt{\text{speed}}$) and lubricant-traction sliding friction (RMS scaling quadratically with speed) – with high-viscosity grease producing friction-dominated, more strongly speed-dependent AE and low-viscosity grease remaining asperity-load-dominated. They further use the coefficient of quartile deviation (CQD) of the time-varying RMS to show that low-viscosity grease produces more strongly fluctuating AE than high-viscosity grease across all frequency sub-bands.

The primary practical challenge is the large data volume generated by continuous AE acquisition at the required sampling rates (>500 kHz), which most studies address through windowed statistical features, interval sampling, or event-triggered recording.

Tandon et al. [45] compare vibration, AE, stator current, and shock pulse for grease contamination detection (silica and ferric oxide particles) in electric motor ball bearings, finding AE most effective for early-stage contamination detection.

Sensor technology. AE sensors are almost exclusively piezoelectric (PZT), available in two configurations with different SNR–bandwidth trade-offs. *Resonant sensors* are tuned to a single resonant frequency (commonly 150 kHz or 300 kHz), providing very high sensitivity in a narrow bandwidth (±15–30 kHz around resonance) but discarding spectral content outside that band. *Wideband sensors* sacrifice sensitivity for a nominally flat response across 100 kHz to 1 MHz, enabling the frequency-domain and time-frequency SP methods reviewed in Section 6; for LCM applications where the informative frequency content is *a priori* unknown or lubrication-condition-dependent, wideband sensors are preferable. Sensor-to-housing coupling (adhesive bond, magnetic mounting, or grease couplant) is a significant source of variability: a poorly bonded sensor can introduce 10–20 dB of additional loss at high frequencies [29], making coupling consistency a prerequisite for any quantitative AE feature.

5.2.2 Vibration

Accelerometers measure mechanical vibration of the bearing housing or adjacent structure. Vibration-based condition monitoring is the most mature and industrially widespread sensing approach for rolling element bearings, with a substantial body of literature on bearing fault detection and remaining useful life estimation [46, 47]. Its application to lubrication condition monitoring is more limited, primarily because lubrication-related processes excite frequency content above 30 kHz, which lies at or beyond the upper limit of typical accelerometers and requires dedicated high-frequency accelerometers or acoustic emission sensors [1].

Vibration signals are sensitive to lubrication conditions through two mechanisms: (1) impulsive excitation from asperity contacts at inadequate Λ, increasing broadband vibration energy, and (2) changes in the rolling element bearing dynamics (stiffness, damping) associated with changes in the lubrication film. Jakobsen et al. [23] demonstrate that vibration statistics and spectral kurtosis are correlated with κ values and can be used to predict lubrication adequacy. Xu et al. [3] show that minimum entropy deconvolution applied to vibration signals enhances weak impulsive events associated with lubricant starvation, improving detectability. Zhang et al. [48] target grease-lubricated bearings specifically, classifying lubrication state into insufficiency, degradation, and contamination from vibration using a deep-learning pipeline rather than handcrafted statistics (see Section 6.6.2).

Sensor technology. Two accelerometer technologies are relevant to LCM, with fundamentally different frequency coverage. *MEMS accelerometers* offer flat response from DC to 1–5 kHz, low cost, and small size, and dominate in IoT-type monitoring nodes – but their upper frequency limit excludes virtually all lubrication-related signal content, which is most informative above 30 kHz. *Piezoelectric (IEPE/ICP) accelerometers* have usable frequency ranges up to approximately one-third of their resonant frequency; standard industrial vibration sensors (resonant frequency 20–30 kHz) are therefore limited to roughly 7–10 kHz – still below the most diagnostically informative range for LCM. *High-frequency piezoelectric accelerometers* with resonant frequencies of 50–150 kHz extend the usable range to 15–50 kHz, providing access to a portion of the lubrication-relevant signal content at the cost of reduced sensitivity at lower frequencies. The consistent finding in the reviewed literature that vibration-based LCM is limited compared to AE and passive ultrasound reflects this hardware constraint: the most LCM-relevant signal content lies above the practical range of standard industrial accelerometers [1].

5.2.3 Ultrasound (>20 kHz)

Ultrasound microphones or wideband accelerometers operating in the frequency range above 20 kHz – distinct from lower-frequency vibration analysis and from active ultrasonic reflectometry – capture the high-frequency acoustic signals generated by friction and wear in the bearing contact zone. These signals are generated passively by the bearing itself and require no external excitation. Ultrasound is sensitive to lubrication changes earlier in the degradation process than lower-frequency vibration because asperity contacts preferentially excite high-frequency content [1]. Low-cost MEMS ultrasonic microphones (e.g. SPH0641LU, bandwidth to 80 kHz) have been demonstrated as an accessible sensor platform for this purpose [49].

Passive ultrasound monitoring is currently used industrially for LCM. Products such as UE Systems Ultraprobe, Sonotec, and AccuTrak employ threshold-based methods that compare the measured ultrasound level against a baseline established under adequate lubrication conditions, instructing relubrication when the level rises a specified number of dB above the reference [50]. Jakobsen et al. [1] demonstrate that more sophisticated processing of passive ultrasound – using band-pass filtering, time-domain features, and regression models – can predict κ values rather than just threshold exceedances.

Sensor technology. Industrial passive ultrasound instruments (handheld or continuous-monitoring) typically employ resonant PZT contact sensors operating in the 20–100 kHz range – a band chosen as a pragmatic balance between SNR advantage over low-frequency structural vibration noise (which improves at higher frequencies) and transmission loss through housing structures (which increases with frequency, as discussed in Section 5.1) [50]. Research-grade instruments use wideband piezoelectric contact sensors covering 20–500 kHz to enable spectral analysis. MEMS ultrasonic transducers are an emerging technology that may enable low-cost, integrated monitoring nodes; their sensitivity and bandwidth characteristics for LCM applications have not yet been systematically evaluated in the reviewed literature.

5.2.4 Audible Sound (20 Hz–20 kHz)

Microphone arrays capture sound emitted by the bearing in the audible range. Asperity contacts, friction, and wear events generate broadband mechanical noise, a portion of which is radiated as airborne sound through the bearing housing; changes in lubrication regime alter the amplitude and spectral distribution of this emission. Georgiev et al. [32] demonstrate that frequency-domain features extracted from microphone array measurements show measurable changes as lubrication levels vary, and propose classification via fuzzy logic or neural networks.

The practical advantage of microphone-based monitoring is its fully non-contact nature, enabling installation without any access to the bearing. The limitation is susceptibility to ambient noise in industrial environments.

Sensor technology. MEMS microphones now achieve noise floors below 30 dB(A) SPL with flat frequency response to 20 kHz, making them suitable for audible-range bearing monitoring at very low cost and in compact form factors. Condenser and electret microphones offer higher sensitivity and lower noise for precision measurements. Signal quality is highly dependent on the acoustic environment of the installation; proximity to the bearing and low ambient noise are prerequisites for useful measurements in industrial settings.

5.2.5 Thermal Monitoring

Temperature sensors or infrared thermography detect heat generation caused by friction and wear inside the bearing, serving as an indicator of lubrication regime. Under adequate EHL lubrication, viscous heat generation is modest; as the bearing transitions to mixed or boundary lubrication, frictional heat increases significantly [34]. Moussa et al. [33] demonstrate a passive thermography approach to bearing condition monitoring in which temperature trend analysis provides early-warning capability.

The primary limitation of thermal monitoring for LCM is its slow response: temperature rise is a downstream consequence of sustained inadequate lubrication. It is therefore more suited to early warning of developing degradation trends than to detection of transient lubrication events.

Sensor technology and emerging approaches. Thermocouples and resistance temperature detectors (RTDs) provide point measurements with response times of seconds to tens of seconds – adequate for trend monitoring but not for transient events. Infrared thermography cameras provide spatially resolved temperature maps without contact but require line-of-sight to the bearing housing surface and are typically used in periodic inspections rather than continuous monitoring. Fibre Bragg grating (FBG) sensors, demonstrated by Janssens et al. [51] and Mohammed et al. [52], offer both thermal and mechanical (vibration) sensing in a single fibre with electrical immunity and multiplexability over a single optical cable – making them attractive for harsh or electrically noisy environments. Wireless temperature sensor nodes with sub-second sampling are commercially available and represent a low-cost option for continuous thermal monitoring in industrial deployments.

5.3 Active Methods

5.3.1 Active Ultrasonic Reflectometry

Active ultrasonic methods transmit high-frequency (∼1–100 MHz) pulses from a piezoelectric transducer into the bearing raceway. At the interface between the raceway and the lubricant film, a portion of the wave is reflected; the reflection coefficient depends on the acoustic impedance of the lubricant film, which is a function of both film thickness and lubricant bulk modulus. Using physics-based acoustic propagation models – most commonly the quasi-static spring-layer model [36] – the film thickness can be recovered from the measured reflection coefficient. Active ultrasonic methods can provide quantitative, real-time measurement of EHL film thickness under operating conditions [37, 38, 53, 54]. Li et al. [16] extend this to the wind turbine pitch bearing, demonstrating film thickness measurement via ultrasonic reflection in a large-scale industrial component. Chen et al. [55] address the data-volume challenge of high-frequency ultrasonic acquisition using sub-Nyquist sampling.

Although transducers can in principle be mounted externally on the bearing housing, the requirement for precise transducer coupling, alignment, and calibration creates practical barriers to deployment on standard industrial bearings that are not present for passive sensing methods [9, 56].

5.3.2 Electrical Methods

Electrical sensing methods treat the ball bearing and lubricant as elements of an electrical circuit. An externally applied electrical signal – voltage or current – probes the electrical properties of the lubricant film, which change with film thickness, lubricant degradation, and debris content. Three variants are used: electrical capacitance [2, 10], electrical impedance [11, 35, 57], and electrical resistance [19]. All require an externally applied electrical signal, making them active methods. Capacitance measurement is primarily sensitive to film thickness; impedance spectroscopy can additionally distinguish lubricant degradation state; resistance measurement detects film breakdown and the onset of metal-to-metal contact. Grease-lubricated bearings present additional modelling challenges due to the heterogeneous dielectric properties of the thickener matrix [26, 58].

5.3.3 Surface Acoustic Wave Sensing

Surface acoustic waves (SAWs) propagate along the surface of elastic materials with rapid depth decay. Piezoelectric probes actively excite SAWs that travel along the raceway surface; as these waves interact with the lubricant film, energy is transferred to the fluid causing wave attenuation, while the lubricant loading also shifts the wave velocity, producing a measurable time delay at a receiving transducer. Both attenuation and time delay can be related to lubrication film thickness and condition [59–61]. The tribo-acoustic tachometer of Tatzgern et al. [60] additionally recovers cage rotational speed from the same signal, providing simultaneous kinematic and lubrication information.

5.3.4 Optical Methods

Optical methods use light-based techniques to either measure lubricant film thickness directly or detect and classify wear debris.

Optical film thickness measurement. Optical interferometry and laser-induced fluorescence (LIF) are the reference methods for EHL film thickness measurement, underpinning much of the fundamental understanding of EHL contact physics [7]. Both require optical access to the contact zone and are restricted to purpose-built tribological test rigs with transparent windows; neither has been demonstrated on standard industrial bearings under field conditions. They are consequently not applicable to online LCM in industrial deployment, but serve as the primary ground-truth measurement methods against which other sensing approaches are calibrated.

Machine-vision debris detection. Machine-vision approaches analyse lubricant debris content using image processing on in-line optical sensors in the lubricant flow path. Liu et al. [40] apply multi-target tracking with Kalman filtering for debris counting and size estimation; Wang et al. [41] use a background-difference extraction step followed by convolutional neural network classification to distinguish debris from bubbles, achieving 91.8% accuracy in a large-diameter, high-flow-rate pipe system. These methods operate on particles in the lubricant flow rather than on the contact zone directly, and can be integrated into oil circulation systems. Unlike interferometry, they are compatible with industrial deployment but provide no film thickness information; their diagnostic value is for contamination and wear monitoring rather than lubrication regime assessment.

5.3.5 Magnetic and Inductive Methods

Magnetic and inductive methods detect metallic wear debris in the lubricant by monitoring perturbations in a magnetic field [14]. An excitation coil generates an alternating magnetic field in or around the lubricant flow path. Ferromagnetic particles perturb the magnetic field primarily through their high permeability, while non-ferrous conductive particles can induce eddy currents that alter the coil impedance. These effects may produce differences in amplitude and phase in the measured signal; in some sensor systems, such differences can be exploited to help distinguish particle types, although reliable material classification and size estimation depend on the sensor design, operating frequency, and signal processing. These methods require access to the lubricant flow and are therefore applicable only to oil-circulating systems; they cannot be used with grease-lubricated or sealed bearings. Shi et al. [62] demonstrate an ultrasensitive impedance-based microsensor for oil condition monitoring applicable to small particle sizes.

6 Signal Processing Methods

This section provides the primary systematic contribution of this review: a taxonomy of signal processing methods applied to LCM of ball bearings, organised by processing family rather than by sensing modality. For each family the method is first described on its own terms – its mathematical basis, assumptions, and outputs – and then its specific applications across sensing modalities are discussed, drawing on the literature surveyed. A master cross-reference matrix linking SP method families to sensing modalities is provided in Table 3 at the end of this section.

The organisation of this section reflects the signal processing chain from raw sensor data to a lubrication condition estimate. Time-domain and frequency-domain methods (Sections 6.1–6.2) operate directly on the measured signal. Time-frequency and decomposition methods (Section 6.3) address non-stationarity. Signal enhancement methods (Section 6.4) improve the signal before feature extraction. Physics-based and model-driven methods (Section 6.5) incorporate lubrication physics to convert processed signals into physically interpretable quantities. Finally, machine learning and data fusion methods (Sections 6.6–6.7) learn mappings from processed features to lubrication condition labels or regression targets. Figure 6 provides an overview of this taxonomy.

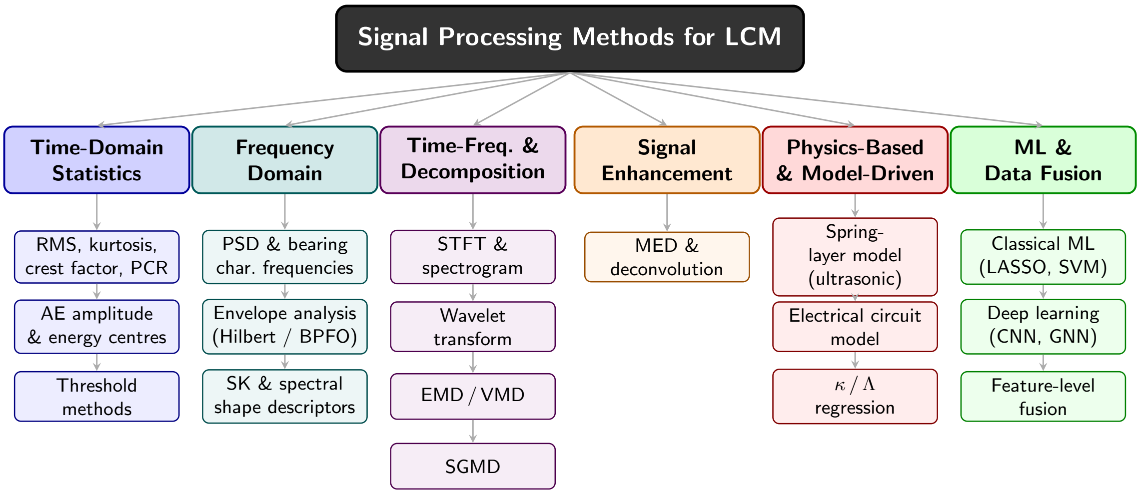

Figure 6: Taxonomy of signal processing method families applied to lubrication condition monitoring of rolling element bearings. Methods are organised by processing family rather than by sensing modality; the master cross-reference matrix (Table 3) maps each family to the sensing modalities for which LCM application has been demonstrated in the reviewed literature.

6.1 Time-Domain Statistical Methods

Time-domain statistical methods extract scalar features from a windowed segment of the raw or bandpass-filtered sensor signal. They are computationally lightweight, robust to non-stationarity within the window, and directly interpretable in physical terms. These properties make them the most practically deployed class of methods in both research and industrial LCM applications.

6.1.1 Standard Statistical Moments and Energy Metrics

The most widely reported features are:

- **Root mean square (RMS):** Reflects the total signal energy over the window and defined as

$$x_{\mathrm{RMS}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2}. \tag{5}$$

  For passive methods, RMS increases monotonically with worsening lubrication in the mixed and boundary regimes due to increasing asperity contact energy. Widely reported for AE [17, 22, 63–65], vibration [1, 3, 23], passive ultrasound [1], and thermal monitoring [33].

- **Kurtosis:** Measures the impulsiveness of the signal relative to a Gaussian distribution (for which $K = 3$) and defined as

$$K = \frac{\frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^4}{\left(\frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^2\right)^2}. \tag{6}$$

High kurtosis indicates sparse, high-amplitude transients such as those produced by asperity micro-contacts or incipient crack impacts. Usgame Sandoval et al. [66] report an ∼8.6-fold increase in AE kurtosis from normal to severe fault in tapered roller bearings. Yoshioka and Shimizu [63] note that under grease lubrication, kurtosis of the raw vibration signal decreases as damage transitions from sparse impulsive AE to a continuous broadband noise floor – a counter-intuitive reversal that illustrates the importance of bearing-specific calibration. Kurtosis applied to spectral representations forms the basis for higher-order spectral analysis of lubrication signals (Section 6.2.3).

- **Crest factor:** peak-to-RMS ratio. Sensitive to impulsive events but tends to decrease as the signal transitions from sparse impulses to continuous broadband noise, similarly to kurtosis. Applied to passive ultrasound and vibration signals for lubrication condition monitoring by Jakobsen et al. [1].

- **Hjorth parameters:** three features derived from successive derivatives of the signal – *activity* (signal variance), *mobility* (ratio of the standard deviation of the first derivative to that of the signal, proportional to mean frequency), and *complexity* (ratio of the mobility of the first derivative to the mobility of the signal, reflecting bandwidth). Jakobsen et al. [1] apply all three to high-pass filtered passive ultrasound signals and find good correlation with κ; combined with linear regression or a shallow neural network they predict κ with good precision. No other paper in the reviewed literature reports Hjorth parameters for LCM.

- **AE amplitude and energy centres (Hou & Zhang):** Rather than a single scalar, Hou and Zhang [17] define the amplitude centre $A_c = \frac{1}{N}\sum A_i$ and energy centre $E_c = \frac{1}{N}\sum E_i$ over a 10-minute window, and form the two-dimensional pair (A_c, E_c). The joint distribution in this space exhibits spatially distinct clusters for boundary, mixed, and EHL lubrication, enabling regime classification without a hard threshold. A speed-dependent weighted power-law regression of the form

$$\Lambda_E = W_A \cdot f_A(A_c, n) + W_E \cdot f_E(E_c, n) \tag{7}$$

  combines the two features – with W_A, W_E linear in shaft speed n – to estimate the film thickness ratio Λ with 89.3% accuracy. This is the most physically grounded time-domain AE approach reported in the reviewed literature.

- **Pulse count rate (PCR):** the number of AE threshold crossings per unit time. Miettinen et al. [30] show that PCR increases monotonically with grease starvation and distinguishes ten grease formulations by thickener type and base-oil viscosity. PCR is the computational minimal AE processing approach and forms the basis of the industrial threshold methods (Section 6.1.2).

- **Ring-down count:** the total number of threshold crossings within a single AE burst event. Usgame Sandoval et al. [66] report a ∼266-fold increase in ring-down count from normal to severe fault – the most sensitive single time-domain feature in their study, outperforming kurtosis and RMS.

- **Coefficient of quartile deviation (CQD):** a robust, distribution-shape-based measure of dispersion in the time-varying RMS, defined as CQD $= (Q_3 - Q_1)/(Q_3 + Q_1)$ for quartiles Q_1, Q_3 of the RMS time series within a window. Zhang et al. [31] use CQD to quantify the fluctuation level of AE signals in cylindrical roller bearings and show that low-viscosity grease produces significantly higher CQD (greater RMS fluctuation) than high-viscosity grease across all analysed frequency sub-bands, consistent with a shift in the dominant AE generation mechanism (Section 5.2.1).

Speed and load normalisation. Time-domain features are sensitive to changes in operating speed and load independently of lubrication condition. Cornel et al. [22] demonstrate this concretely: the AE amplitude–λ relationship follows distinct curves at 500 rpm and 3000 rpm, making a single amplitude threshold insufficient for regime classification across speeds. This confounding effect is addressed by normalising features against a speed-dependent or load-dependent baseline; Hou and Zhang [17] explicitly incorporate speed as a predictor in their film-thickness regression. Without such normalisation, a feature increase caused by a speed change is indistinguishable from a lubrication-related change, which limits the applicability of fixed-threshold industrial methods to constant-speed machinery.

6.1.2 Threshold Methods

The simplest operational approach is to compare a time-domain feature – most commonly the ultrasound dB level or AE RMS – against a threshold established during a reference measurement under known adequate lubrication. If the feature rises a specified number of dB above the reference, relubrication is recommended [50]. This is the method employed by industrial instruments such as UE Systems Ultraprobe and Sonotec's handheld devices. The advantages are simplicity, interpretability, and no requirement for labelled training data. The critical disadvantage is that the threshold carries no information about operating condition: a speed or load change will trigger a false alarm, and gradual lubrication degradation that produces a slow trend may not exceed the threshold until damage is well developed.

6.2 Frequency-Domain Methods

Frequency-domain methods transform the time-domain signal into a spectral representation, enabling identification of narrowband features associated with periodic bearing kinematics superimposed on broadband tribological noise.

6.2.1 Power Spectral Density and Bearing Characteristic Frequencies

The Fourier transform and power spectral density (PSD) reveal the distribution of signal energy across frequency. In the context of LCM, two types of spectral content are of interest: (1) broadband increases in spectral energy associated with increased asperity contact under inadequate lubrication, and (2) narrowband peaks at bearing characteristic frequencies – the ball pass frequency of the outer race (BPFO), ball pass frequency of the inner race (BPFI), ball spin frequency (BSF), and fundamental train frequency (FTF) – which indicate rolling element excitation of resonances in the bearing structure. Frequency-domain features carry lubrication-related information beyond what is captured by time-domain statistics alone.

6.2.2 Envelope Analysis

Envelope analysis demodulates a high-frequency carrier signal – such as a structural resonance excited by bearing impacts – to extract the low-frequency amplitude modulation produced by periodic contact events [46]. The approach consists of: (1) bandpass filtering around a resonance frequency to isolate the carrier, (2) computing the analytic signal via the Hilbert transform to obtain the instantaneous envelope, and (3) computing the FFT of the envelope (the envelope spectrum) to reveal modulation frequencies. In the LCM context, envelope analysis is most naturally applied to AE signals above 100 kHz, where lubrication-induced asperity contact events modulate the high-frequency carrier. Kestel et al. [61] apply AE envelope analysis as part of a multi-sensor LCM system.

6.2.3 Higher-Order Spectral Statistics and Spectral Shape Descriptors

Higher-order spectral statistics and shape descriptors characterise the spectral distribution beyond its energy content. Whether computed on a one-dimensional amplitude spectrum or per frequency bin across the time dimension of a two-dimensional STFT spectrogram, these are fundamentally shape analyses: kurtosis measures impulsiveness and peakedness, skewness measures asymmetry, flatness the energy concentration relative to white noise, and center frequency, RMS frequency, and bandwidth-related measures the location and spread of spectral energy.

Jakobsen et al. [23] apply spectral kurtosis (SK) to vibration signals from grease-lubricated ball bearings under alternating lubrication conditions, with speed and temperature varied to span κ from approximately 0.2 to 3.0. SK is computed as the normalised fourth moment of the STFT coefficient distribution at each frequency bin across successive time slices, reducing the two-dimensional spectrogram to a one-dimensional kurtosis profile. Band energy extracted from a lubrication-sensitive kurtogram band (7500–12000 Hz) is correlated directly with κ using Pearson and Spearman coefficients; values of $r = -0.76$, $\rho = -0.84$ are reported during the first hot-run stage ($\kappa < 0.5$). The kurtogram band energy serves as the lubrication condition indicator itself, not as a selector for downstream envelope analysis.

Zhang et al. [67] take a complementary approach, computing 15 spectral shape descriptors (features F10–F24 in their Table 1) from vibration signals in the 0–3 kHz and 10–12.8 kHz bands and from AE signals in the 20–40 kHz band. These include dominant frequency, band-specific energy, center frequency, RMS frequency, spectral mean and standard deviation, spectral flatness, frequency-domain skewness and kurtosis, and normalised bandwidth. Features are selected by monotonicity and time-correlation criteria and combined with time-domain statistics as node embeddings in an MTAD-GAT model for lubrication degradation detection; temperature receives no spectral processing. The kurtosis term in Zhang et al. (F15) is the fourth standardised moment of the one-dimensional amplitude spectrum – a global scalar shape statistic – whereas Jakobsen et al.'s SK is a per-frequency-bin statistic computed over the time dimension of the STFT; both are instances of applying kurtosis to a spectral representation.

6.2.4 Fractional-Octave Filter Banks

Rather than a single broadband metric, fractional-octave filter banks decompose the signal into a set of frequency bands following a geometric frequency progression. Renhart et al. [65] apply a 1/12-octave filter bank (spanning approximately 50–425 kHz) to AE signals from journal bearing segments. Within each band and time window, they compute the composite parameter $P_5 = x_{\mathrm{RMS}} \cdot K$ – the product of RMS and kurtosis – which is sensitive to transient impulsive events that are simultaneously energetic and non-Gaussian. The approach reveals mechanistic discrimination unavailable from broadband time-domain statistics: the 86.8 kHz band is dominant during adhesive wear at low speeds, while the 206–232 kHz bands increase at higher speeds and are attributed to particle-driven wear This frequency-resolved mechanistic discrimination is a key advantage of spectral processing for LCM.

6.2.5 Multi-Band Hit-Rate and Wear Particle Index

The Delft University group [43] employ a three-band sensor array (40–100 kHz, 95–180 kHz, 180–580 kHz) for AE monitoring of large-scale low-speed slew bearings. An AE hit is recorded when the signal in a given band exceeds a threshold with a minimum SNR of 20 dB, and the hit rate is normalised by loading cycles to remove the effect of varying test duration. A Wear Particle Index (WPI) derived from the normalised hit rate shows linear correlation with abrasive particle mass fraction in grease. Crucially, non-abrasive (soft) particles produce negligible additional hits, demonstrating that AE hit-rate is selective for hardness contrast rather than total particle mass – a practically important limitation for contamination monitoring.

A related but distinct use of sub-band decomposition is mechanism attribution rather than fault or wear detection. Zhang et al. [31] divide the 1–200 kHz AE spectrum into 20 kHz-wide sub-bands and, within each, fit the RMS-versus-speed trend and its coefficient of determination (R^2) to identify whether the asperity-load ($\sqrt{\text{speed}}$) or sliding-friction (speed^2) mechanism dominates. This reveals that the two AE generation mechanisms identified in Section 4 are not uniformly distributed across frequency: for low-viscosity grease the lower sub-bands trend closer to the square-root relationship while higher sub-bands trend closer to quadratic, whereas for high-viscosity grease the quadratic trend holds across all sub-bands.

6.3 Time-Frequency and Decomposition Methods

Time-frequency methods represent signal energy jointly in time and frequency, addressing the non-stationarity that characterises LCM signals arising from variable operating conditions or transient lubrication events.

6.3.1 Short-Time Fourier Transform

The STFT divides the signal into short, overlapping windows and computes the FFT of each, yielding a spectrogram. The time-frequency resolution is governed by the window length (longer windows yield better frequency resolution but poorer time resolution). For LCM, the spectrogram is used to track the evolution of spectral features over time and to visualise non-stationary changes associated with lubrication degradation. The STFT is also used within the SK framework as the basis for computing the kurtogram [23].

6.3.2 Wavelet Transform

The continuous wavelet transform (CWT) provides time-frequency resolution that adapts to the local signal frequency through a scaling parameter: short wavelets at high frequencies and long wavelets at low frequencies, giving better time resolution where signals are fast-varying and better frequency resolution where they are slow-varying. This property is well-suited to AE bursts, which are transient and broadband. Renhart et al. [65] apply the CWT with a Morse (generalised Morse) wavelet to individual AE bursts triggered by the P_5 criterion, producing amplitude scalograms that reveal a two-stage within-event mechanism: an initial high-frequency asperity impact followed by a lower-frequency relaxation. Adaptive wavelet denoising – in which the wavelet decomposition threshold is optimised by a kurtosis-entropy criterion – is a further variant applicable to bearing signals, though no reviewed study applies this specifically to lubrication condition monitoring.

The discrete wavelet transform (DWT) and its multi-resolution analysis framework enable simultaneous extraction of features at multiple frequency scales, which is in principle applicable to vibration signals for LCM, though no reviewed study reports DWT specifically for lubrication condition monitoring of vibration signals.

6.3.3 Signal Decomposition Methods

Empirical mode decomposition (EMD) and its variants – ensemble EMD (EEMD) and variational mode decomposition (VMD) – adaptively decompose a signal into intrinsic mode functions without requiring a predefined basis, making them suited to non-linear and non-stationary bearing signals [8]. Although these methods are used in general bearing condition monitoring, no study in the reviewed LCM literature applies EMD or VMD specifically to lubrication condition estimation, leaving them as a potential avenue for future work.

Symplectic geometry mode decomposition (SGMD) is a decomposition method based on the symplectic geometry of the phase space of the signal's trajectory matrix. Yu et al. [39] apply SGMD to signals from inductive debris sensors, extracting debris-particle signatures buried in noise and harmonic interference. Pan et al. [68] provide the methodological foundation. The method offers improved separation of overlapping frequency components compared to EMD-based methods in specific application scenarios.

6.4 Signal Enhancement Methods

Signal enhancement methods improve the signal-to-noise ratio or impulsiveness of the raw sensor data prior to feature extraction, without reducing the signal to a scalar feature.

6.4.1 Minimum Entropy Deconvolution

Minimum entropy deconvolution (MED) designs a finite impulse response (FIR) filter that maximises the kurtosis (equivalently, minimises the entropy) of the filter output, thereby recovering the sparse impulsive excitation source from a convolved, noisy observation. In the LCM context, the impulsive sources are the transient mechanical impacts associated with asperity contacts or rolling element passage over roughened raceways under starved lubrication. Xu et al. [3] demonstrate that MED applied to vibration signals significantly enhances the detection of lubrication starvation-related impulses, enabling spectral centroid and envelope features to detect starved conditions that are not detectable in the unprocessed signal. The method assumes that the source signal is sparse (impulsive), which may not hold for all lubrication fault types. Multipoint optimal minimum entropy deconvolution adjusted (MOMEDA) and its variants extend MED to multichannel and periodic-source scenarios.

6.5 Physics-Based and Model-Driven Methods

Physics-based methods incorporate lubricant or structural mechanics models to convert processed signals into physically interpretable quantities such as film thickness or lubrication regime.

6.5.1 Quasi-Static Spring-Layer Model for Ultrasonic Reflectometry

Active ultrasonic reflectometry measures the reflection coefficient R of an ultrasonic pulse at the raceway–lubricant interface. For a thin lubricant film (film thickness much less than the acoustic wavelength), the spring-layer model relates R to the film stiffness K^*:

$$R = \frac{Z_2 - Z_1 - iZ_1Z_2/\omega K^*}{Z_2 + Z_1 + iZ_1Z_2/\omega K^*} \tag{8}$$

where Z_1, Z_2 are the acoustic impedances of the raceway and lubricant materials, ω is the angular frequency of the ultrasonic pulse, and K^* is the complex stiffness of the lubricant layer related to film thickness h and lubricant bulk modulus E^* by $K^* = E^*/h$ [15, 36]. This model enables direct, calibrated film thickness estimation from the measured reflection coefficient. Validation against Hamrock–Dowson predictions is reported in [37, 38, 53]. Phase-based algorithms that improve accuracy for thin films are described in [56]. Sub-Nyquist sampling to reduce data acquisition requirements while retaining film thickness accuracy is demonstrated by Chen et al. [55]. Li et al. [16] extend the spring-layer model to the curved contact geometry of wind turbine pitch bearings using ray-modelling corrections for transducer curvature and surface geometry.

6.5.2 Electrical Circuit Models

For electrical sensing methods (capacitance, impedance, resistance), physics-based circuit models relate the measured electrical signal to the lubricant film thickness. The bearing is modelled as a parallel combination of capacitive and resistive elements whose values depend on the film thickness, lubricant dielectric constant, and Hertzian contact geometry. Cen et al. [10] and Becker-Dombrowsky et al. [11] review the capacitance and impedance formulations respectively. Maruyama et al. [35] apply the impedance model to practical deep-groove ball bearings under operating conditions. Two approaches to model calibration are reported: purely physics-based models using known material properties, and empirical calibration tests using reference lubricants to derive a relationship between the measured signal and Hamrock–Dowson-predicted film thickness [10, 57].

6.5.3 Kappa and Lambda as Regression Targets

A model-driven approach for passive sensing methods uses κ or Λ – computed from the Hamrock–Dowson model and operating parameters – as the regression target for data-driven models trained on sensor features. This provides a physically grounded target variable linked to ISO standards without requiring invasive lubricant sampling. Jakobsen et al. [1] combine the Hjorth parameters defined in Section 6.1.1 with LASSO regression and shallow neural networks to predict κ from passive ultrasound signals, directly bridging physics-based lubrication modelling and data-driven signal processing.

6.6 Machine Learning and Deep Learning

Data-driven methods learn mappings from handcrafted or raw signal features to lubrication condition labels or continuous condition estimates. They can be organised into classical machine learning (operating on handcrafted features) and deep learning (operating on raw signals or spectrograms).

6.6.1 Classical Machine Learning

Classical machine learning methods – support vector machines (SVM), random forests, logistic regression, and regularised linear regression (LASSO) – are applied to vectors of handcrafted features extracted by the time-domain, frequency-domain, or time-frequency methods described above. Jakobsen et al. [1] demonstrate that LASSO regression on passive ultrasound features achieves good κ prediction while providing interpretable feature weights.

6.6.2 Deep Learning

Deep learning methods operate on raw signals or intermediate representations (spectrograms, graph adjacency matrices) and learn features jointly with the classification or regression task. Key architectures applied to LCM include:

- **Convolutional neural networks (CNNs):** applied to lubricant debris images for particle classification [41]. Zhang et al. [64] apply a vision-language large language model (PaLiGemma) to images of AE time-domain waveforms for automated wear detection, reporting higher classification accuracy than conventional threshold methods.

- **Graph neural networks (GNNs):** Zhang et al. [67] represent multi-sensor data (vibration, AE, temperature) as a graph and apply graph convolutional operations to detect lubrication degradation, leveraging sensor correlations that are not captured by independent feature extraction.

- **Generative adversarial domain transfer (GANs):** Zhang et al. [48] address the zero-labelled-fault-data problem directly with a multiscale residual and semantic-consistency CycleGAN (MSRS-CycleGAN) that translates labelled multi-state laboratory vibration signals into field-style signals, using healthy-only field vibration as the unlabelled target domain. Multiscale residual convolutional blocks improve sensitivity to the broadband, high-frequency signatures characteristic of lubrication anomalies (as opposed to the narrowband impulsive signatures of mechanical damage), and a semantic-consistency loss preserves the insufficiency/degradation/contamination state distinction during lab-to-field style transfer. A downstream autoencoder trained only on healthy field data performs anomaly detection, followed by CNN classification of the generated field-style anomaly signals into lubrication states. This is the only reviewed method that produces usable lubrication-anomaly training data for a field deployment with no labelled field-fault examples at all.

A consistent challenge for deep learning in LCM is the scarcity of labelled training data from real bearing degradation under controlled lubrication conditions. Controlled accelerated tests are expensive to run, and publicly available LCM datasets with varied lubrication conditions and sensing modalities are not currently available. Generative domain transfer [48] is an emerging response to this challenge for the specific case of lab-to-field transfer, though it remains distinct from the self-supervised and foundation-model directions discussed in Section 8.6, which target representation learning from unlabelled data rather than synthesis of labelled examples in a new domain.

6.7 Multi-Sensor Data Fusion

Multi-sensor fusion combines information from multiple sensing modalities to obtain lubrication condition estimates more accurate or robust than any single modality alone.

6.7.1 Feature-Level Fusion

Feature-level (early) fusion concatenates feature vectors from multiple sensors into a single feature vector before the classification or regression step. Zhang et al. [67] provide the clearest example in the lubrication condition monitoring literature: vibration, AE, and temperature signals are represented as a graph in which nodes correspond to sensing modalities and edges encode their relationships through a priori domain knowledge; a graph attention network trained on this structure detects lubrication degradation, and the multi-sensor system outperforms any single-modality baseline. The graph-attention architecture is notable because it learns which modality combinations are most informative for a given operating condition, rather than weighting all sensor contributions equally.

6.8 Master Cross-Reference Matrix

Table 3 summarises the application of SP method families to each sensing modality, as found in the reviewed literature. A filled cell indicates the combination has been demonstrated for LCM in at least one reviewed study.

SP Method Family	Vibration	Acoustic Emission	Ultrasound (passive)	Sound / Microphone	Thermal	Active Ultrasonic	Electrical	Magnetic / Oil
Time-domain statistics	✓	✓	✓		✓			✓
Threshold methods		✓	✓					
PSD / Bearing char. freqs	✓	✓	✓	✓				
Envelope analysis		✓						
Spectral shape & kurtosis	✓	✓						
Octave filter bank		✓						
STFT / Spectrogram	✓							
Wavelet transform		✓						
Signal decomposition (SGMD)								✓
MED / Deconvolution	✓							
Spring-layer model						✓		
Electrical circuit model							✓	
κ/Λ regression target	✓	✓	✓					
Classical ML (SVM, LASSO)	✓	✓	✓	✓				✓
Deep learning (CNN, GNN, GAN)	✓	✓			✓			
Multi-sensor fusion	✓	✓			✓			✓

Table 3: Master cross-reference matrix of signal processing method families and sensing modalities for LCM of ball bearings. A ✓indicates the combination has been demonstrated in at least one study included in this review.

7 Comparative Analysis

This section synthesises the evidence from Sections 5 and 6 into a structured comparison of sensing and signal processing method combinations along four dimensions: sensitivity to specific lubrication fault types, sensitivity to operating conditions, practical deployment requirements, and technology maturity. The goal is not to identify a single best method – no such universal optimum exists, since the optimal combination is application-specific – but to characterise the trade-offs that determine which combination is appropriate for a given deployment context.

7.1 Sensitivity to Lubrication Fault Type

Three principal lubrication fault types are of interest in bearing LCM: (1) *lubricant starvation* (insufficient quantity), (2) *viscosity degradation* (thermal or oxidative aging reducing viscosity), and (3) *contamination* (external or wear debris in the lubricant). These faults produce partially distinct signal signatures, which determines which SP methods are most applicable to each. Table 4 summarises the reported sensitivity of each method combination to each fault type.

Sensing method	SP method	Starvation	Viscosity degradation	Contamination
Vibration	Time-domain (RMS, kurtosis)	Moderate [3, 23]	Low–Moderate [23]	
Vibration	MED + envelope	High [3]		
Vibration	Spectral kurtosis + κ regression	Moderate–High [23]	Moderate [23]	
AE	Time-domain (RMS, PCR)	High [17, 22, 30]	Moderate [22]	Moderate [30]
AE	Octave filter bank (P_5)	High [65]		High [65]
AE	Multi-band hit-rate + WPI			High (abrasive only) [43]
Passive ultrasound	Threshold (dB)	Moderate [50]		
Passive ultrasound	Spectral kurtosis + κ regression	Moderate–High [1]		
Thermal	Time-domain (trend)	Moderate (slow response) [33]		
Active ultrasonic	Spring-layer model	High (film thickness) [37, 38]	High (film thickness) [37]	
Electrical	Circuit model	High (film thickness) [10, 11]		
Magnetic / inductive	Mode decomp. + pulse			High (ferrous debris) [39]

Table 4: Reported sensitivity of sensing and signal processing method combinations to three principal lubrication fault types. All ratings are based on direct experimental evidence in the reviewed literature. Empty cells indicate the combination has not been assessed for that fault type in the reviewed literature. Parenthetical notes indicate the specific quantity or condition reported.

Several cross-cutting observations emerge from Table 4. First, no single passive non-invasive method combination provides high sensitivity to all three fault types simultaneously. AE with time-domain features performs best across starvation, viscosity degradation, and contamination, but the sensitivity to contamination is selective for abrasive hard particles (as demonstrated by the Delft group for AE hit-rate). Active ultrasonic and electrical methods provide quantitative film thickness estimates and are therefore inherently sensitive to both starvation and viscosity degradation through their direct relationship with h_m, but they do not directly detect debris contamination. Magnetic and inductive methods specifically target ferrous debris and provide complementary coverage for contamination monitoring, but require access to the lubricant flow and are inapplicable to grease-lubricated or sealed bearings.

Second, passive methods are most sensitive to starvation – the fault type that generates the greatest increase in asperity contact energy – and least sensitive to viscosity degradation. Viscosity degradation reduces κ and shifts the lubrication regime boundary, but whether the operating point crosses into mixed or boundary lubrication depends on the initial viscosity margin; bearings operating with a large film thickness reserve may remain in the EHL regime even with significantly degraded lubricant, producing little change in passive sensor signals.

Third, the temporal response of thermal monitoring is qualitatively different from acoustic methods: temperature rise is a consequence of sustained elevated friction and therefore lags the onset of inadequate lubrication by minutes to hours, whereas AE and vibration respond on the timescale of individual asperity contacts (microseconds to milliseconds). Thermal monitoring is therefore better suited to detecting the existence of inadequate lubrication conditions than to detecting their onset or intermittent occurrence.

7.2 Sensitivity to Operating Conditions

All passive sensing methods are confounded by variations in operating speed, load, and temperature to varying degrees. This confounding is the primary practical challenge for LCM in industrial machinery that operates at variable conditions. Table 5 summarises the known dependencies and compensation strategies reported in the literature.

Method	**Speed dependence**	**Load dependence**	**Temperature dependence**	**Compensation strategy**
Vibration time-domain	High [23]	High [23]		Speed/load normalisation; κ as target [23]
Vibration spectral kurtosis + κ regression	High [23]	High [23]		κ target encodes speed, load, viscosity via Hamrock–Dowson model [23]
AE time-domain	High [17]	Moderate [22]		Speed-weighted features [17]
AE frequency domain (octave, SK)				No compensation strategy reported in reviewed literature
Passive ultrasound threshold	High [50]			Reference measurement at same operating point [50]
Passive ultrasound κ regression	Moderate [1]	Moderate [1]		κ target implicitly encodes operating conditions; most principled passive compensation strategy [1]
Thermal monitoring			N/A (temperature is the signal)	Thermal model as baseline [33]
Active ultrasonic (film)	Low (direct measurement) [37]	Low [37]	Moderate (lubricant viscosity varies) [37]	EHL model with temperature correction [37]
Electrical (capacitance/impedance)	Low (direct measurement) [10]	Low [10]	Moderate (dielectric constant varies) [11]	Temperature-corrected circuit model [10]

Table 5: Operating condition dependence of sensing and signal processing method combinations for LCM, with reported compensation strategies.

The fundamental difficulty for passive methods is that the same signal change – an increase in RMS, AE energy, or ultrasound dB level – can be produced by an increase in speed, load, or temperature as well as by worsening lubrication. Active methods (electrical, ultrasonic reflectometry) provide more direct film thickness measurements and are inherently less speed- and load-dependent, though they remain sensitive to temperature through lubricant viscosity. The κ regression approach of Jakobsen et al. [1, 23] is noteworthy as a principled strategy for passive methods: by training regression models with κ as the target – a variable that already encodes the speed, load, and viscosity dependence through the Hamrock–Dowson and ISO 281 models – the learned model implicitly compensates for these operating condition effects.

7.3 Practical Deployment Requirements

Table 6 compares sensing and signal processing method combinations on practical deployment characteristics relevant to industrial implementation.

Sensing method	Bearing modif.	Sensor installation constraints	Sampling rate	Real-time capable	Calibration / data requirements
AE	None	Flat surface, direct structural path to bearing bore; avoid joints and material changes; bond or clamp required for high-freq. coupling	>500 kHz	Partial (features only)	Baseline measurement under known lubrication condition [50]; labelled data for ML
Vibration	None	Standard industrial accelerometer mount (stud, adhesive, magnet); location affects frequency content but less critically than AE	≤50 kHz	Yes	No baseline required; labelled data for ML
Passive ultrasound	None	As for AE, but less sensitive to exact path geometry at 20–100 kHz; industrial instruments are handheld or clamp-on	20–500 kHz	Yes	Baseline reference at known good lubrication [50]; labelled data for regression
Sound / microphone	None	Non-contact; proximity (<0.5 m) and low ambient noise critical; high-freq. content lost beyond 20 kHz in air	≤50 kHz	Yes	No baseline; labelled data for ML
Thermal	None	Housing surface or non-contact IR; robust to mounting position; slow response renders exact location unimportant	Low (Hz)	Yes	No; threshold or trend analysis
Active ultrasonic	Transducer coupling	Requires permanent or clamp-on transducer with controlled couplant; alignment critical; curved surfaces need geometric correction [16]	>1 MHz	Partial	Calibration reference signal essential [9]; no labelled data (model-based)
Electrical	Electrical path through bearing	Requires electrical isolation of bearing from frame; brush or slip-ring contacts to rotating shaft; not retrofittable in standard machines	kHz–MHz	Partial	Calibration reference [10]; no labelled data (model-based)
Magnetic / inductive	Access to lubricant flow	Sensor in or around lubricant line or sump; not applicable to grease-lubricated sealed bearings	kHz	Yes	Particle count baseline; no labelled data

Table 6: Practical deployment characteristics of sensing methods for LCM. "Bearing modification" refers to structural changes to the bearing or housing beyond sensor attachment. "Real-time capable" refers to whether the SP method can produce condition estimates at bearing rotational rates without offline computation. Installation constraints reflect both sensor coupling requirements and signal propagation considerations (Section 5.1).

Passive non-invasive methods (AE, vibration, ultrasound, sound, thermal) share the critical practical advantage that they require no bearing modification and can in principle be retrofitted to bearings in service. The trade-off is that their signals carry lubrication information mixed with other sources (machine vibration, electromagnetic interference, structural resonances), requiring more sophisticated SP to extract the lubrication-relevant component. Active and invasive methods provide cleaner, more direct lubrication information at the cost of installation complexity and, in some cases, bearing modification.

The installation constraints in Table 6 highlight a practical hierarchy among passive non-invasive methods. AE is the most installation-sensitive: the frequency-dependent attenuation described in Section 5.1 means that signal quality degrades significantly if the sensor is separated from the bearing by bolted joints, material transitions, or long structural paths. A sensor poorly positioned relative to the bearing can render high-frequency AE content undetectable regardless of its nominal sensitivity. Passive ultrasound at 20–100 kHz is more tolerant of imperfect mounting, which partly explains its wider industrial adoption. Vibration accelerometers, operating at yet lower frequencies, are the most robust to mounting position variations. Thermal sensors, operating at DC, are entirely insensitive to structural transmission path effects. Among passive methods, AE imposes the highest data-rate requirement (>500 kHz sampling) due to the high-frequency nature of the emitted waves, which creates storage and computation challenges for continuous real-time monitoring. The two-dimensional amplitude-energy framework of Hou and Zhang [17] demonstrates a computationally demanding offline analysis; real-time implementations would require significant simplification or dedicated hardware.

7.4 Technology Maturity

Table 7 characterises the technology readiness of each method combination on a three-level scale: *Research* (demonstrated only in controlled laboratory conditions), *Prototype* (demonstrated on industrial components or in field trials, but not in commercial products), and *Industrial* (available in commercial products or deployed in regular industrial use).

Sensing method	SP method	Maturity	Representative evidence
Passive ultrasound	dB threshold	Industrial	UE Systems, Sonotec, AccuTrak products [50]
Vibration	Time-domain + frequency-domain (general bearing CM; LCM via threshold)	Industrial	Widely deployed; Brüel & Kjær, SKF, Schaeffler products
Active ultrasonic	Spring-layer model	Prototype	[16, 37, 38]
Electrical (capacitance/impedance)	Circuit model	Prototype	[10, 11, 57]
AE	Time-domain (RMS, PCR)	Prototype	[17, 22, 30]
Vibration	κ regression (LASSO, NN)	Research	[23]
Passive ultrasound	κ regression	Research	[1]
AE	Octave filter bank + P_5	Research	[65]
AE	GNN / multi-sensor fusion	Research	[67]
Sound / microphone	Frequency features + ML	Research	[32]
SAW	Spring-layer model	Research	[59, 60]

Table 7: Technology maturity of sensing and signal processing method combinations for LCM. Maturity levels: *Industrial* – commercially deployed; *Prototype* – demonstrated on industrial components or in field tests; *Research* – demonstrated in controlled laboratory studies only. Note: the *Industrial* maturity of vibration refers to general bearing condition monitoring; LCM-specific vibration deployment is limited to simple threshold methods not identified as a dedicated LCM product in the reviewed literature.

A striking observation is the large gap between the industrial deployment of passive ultrasound threshold methods and the research-stage status of virtually all more sophisticated SP approaches. The threshold method – despite its well-documented limitations (speed-load confounding, no degradation modelling) – is the only passive non-invasive LCM method in routine commercial deployment. All regression, ML, and physics-informed SP methods remain at the research stage, with none yet validated in a full industrial deployment at scale. This gap constitutes a primary motivation for the research direction described in Section 8.

7.5 Summary of Comparative Findings

The comparative analysis reveals a structural inversion at the core of the LCM field: the sensing methods most accessible for industrial deployment – passive, non-invasive, retrofittable without bearing modification – are supported by the least mature signal processing, while the methods with the most physically grounded and quantitatively capable SP (active ultrasonic reflectometry, electrical circuit models) impose installation barriers that restrict them to controlled test environments or high-value dedicated assets. Industrial deployment of passive LCM is consequently dominated by threshold methods developed in the early 2000s, while the diagnostically superior SP approaches – physics-informed κ regression, multi-sensor GNN fusion – remain at the research stage, validated only under constant operating conditions. Closing this gap requires SP methods that are simultaneously physically interpretable, robust to operating condition variation, and deployable on standard industrial hardware without bearing modification – the challenge that defines the research gaps identified in Section 8.

8 Research Gaps and Future Directions

The comparative analysis in Section 7 reveals several gaps in the current state of LCM research that are specifically related to signal processing. This section identifies six gaps that represent the most consequential barriers to advancing passive, non-invasive LCM from research to industrial practice, and proposes future research directions for each.

8.1 Absence of Standardised Benchmark Datasets

The reviewed literature presents results from a wide variety of experimental setups: different bearing types, sizes, operating speed and load ranges, lubricant types, fault induction methods, and measurement configurations. No standardised publicly available dataset exists for LCM analogous to the CWRU, FEMTO-ST, or MFPT datasets that have enabled systematic comparison of fault detection methods in the broader bearing diagnostics community. As a result, no reviewed study is directly comparable to another in quantitative terms, and the relative merits of different SP approaches cannot be rigorously assessed from the literature alone.

Future direction: A standardised LCM benchmark dataset should include multiple passive sensing modalities (AE, vibration, passive ultrasound) recorded simultaneously with an active reference method (electrical capacitance, active ultrasonic film thickness, or friction measurement) that provides ground-truth Λ values or Stribeck curve data. The dataset should span the boundary, mixed, and EHL regimes under systematically varied speed, load, and temperature, with both oil and grease lubrication and at least one contamination condition. Making such a dataset publicly available would substantially accelerate research in both SP methods and data-driven approaches.

8.2 Speed- and Load-Invariant Signal Processing

Virtually all SP methods in the reviewed literature are validated under constant or near-constant speed and load conditions. Real industrial machinery operates under continuously varying loads and speeds, and no SP method has been demonstrated to provide reliable LCM under realistically varying industrial operating conditions. The κ regression approach of Jakobsen et al. [1] is the most principled attempt at operating-condition-aware processing, but it is validated only over a limited range of discrete operating points and does not address continuous variation.

Future direction: SP methods for LCM under variable operating conditions should be developed and validated. Promising approaches include: physics-informed normalisation that uses the Hamrock–Dowson model to compute a condition-dependent baseline feature level; order tracking to convert time-domain features to shaft-angle domain, removing speed dependence from periodic signal components; and operating-condition-aware machine learning models that take speed and load as conditioning inputs or use contrastive learning to learn operating-condition-invariant representations.

8.3 Grease Lubrication Monitoring Is Underexplored

The majority of quantitative SP results in the reviewed literature concern oil-lubricated bearings under controlled laboratory conditions. Grease lubrication – which is employed in the majority of industrial rolling element bearings – presents additional complexity: grease degrades through base-oil bleed-out, thickener structural breakdown, thermal degradation, and contamination while the relationship between grease condition and measurable signal is less well characterised than for oil [24, 25, 27]. Miettinen et al. [30] and Scheeren et al. [44] are among the few studies that specifically characterise passive AE for grease lubrication.

Future direction: Controlled studies systematically varying grease degradation state (base-oil bleed-out, aging time, thickener degradation, contamination type and concentration) while simultaneously measuring passive sensor signals and reference quantities (film thickness by electrical or optical methods, grease rheological properties by in-situ rheometry, or friction measurements for Stribeck curve data) are needed. The specific contribution of thickener structure to AE and vibration signals – separate from base-oil film-forming effects – has not been isolated in the reviewed literature.

8.4 Maintenance Decision Integration

The LCM pipeline in Figure 1 includes a maintenance decision step (relubrication timing and quantity) downstream of the condition assessment. In the reviewed literature, this integration is almost entirely absent: studies terminate at condition assessment (classifying lubrication regime or estimating Λ/κ) without translating the estimate into a relubrication decision. The industrial threshold methods (UE Systems, Sonotec) effectively collapse the assessment and decision steps into a single threshold, but this approach is overly conservative and does not account for degradation rate or operating conditions. No reviewed SP method produces an output directly usable as a relubrication decision criterion – for example, lubrication amount for restoring adequate lubrication conditions, a predicted time-to-failure under the current lubrication condition, or an optimal relubrication interval estimate.

Future direction: Prognostic models that extend the LCM output from current condition assessment to future condition prediction should be developed. Remaining useful life (RUL) estimation for lubrication degradation – distinct from mechanical fault RUL – is an open problem. Approaches could include survival models as in [69] conditioned on the lubrication condition estimate, continuous feedback based upon applied lubrication, or reinforcement learning-based relubrication policies that optimise between the costs of under- and over-lubrication.

8.5 Multi-Sensor Fusion for Passive, Non-Invasive Sensing

Multi-sensor fusion is demonstrated in the reviewed literature primarily in controlled laboratory settings with multiple passive sensors on the same test rig [22, 67]. No reviewed study demonstrates multi-modal passive non-invasive LCM on a bearing in an actual industrial machine, where sensor placement is constrained, ambient noise is high, and the sensing system must operate autonomously. The complementarity of AE, vibration, and passive ultrasound for lubrication fault detection is argued in multiple papers but not quantified in a unified experimental framework with a common ground-truth reference.

Future direction: A systematic experimental study combining AE, vibration, passive ultrasound, and temperature sensing simultaneously on an industrial bearing under realistic operating conditions, with an active film thickness reference, would establish the quantitative benefit of multi-modal fusion for LCM. Early and late fusion architectures should be compared against each sensor modality alone using consistent fault conditions and evaluation metrics.

8.6 Self-Supervised and Foundation Model Approaches for Data-Scarce LCM

A fundamental barrier to data-driven LCM is the scarcity of labelled training data. Generating controlled bearing degradation data under varied lubrication conditions is expensive, time-consuming, and results in small, application-specific datasets. Self-supervised learning (SSL) methods pre-train representations on large amounts of unlabelled passive, non-invasive data and fine-tune on small labelled sets, offering a route to high-accuracy models with reduced labelled data requirements. The TF-C framework [70] and time-series SSL survey [71] demonstrate that SSL representations transfer across mechanical fault detection tasks, but direct application to LCM – where the label is a continuous lubrication condition variable rather than a discrete fault class – has not been demonstrated.

A related but distinct response to data scarcity is generative domain transfer rather than representation pre-training: Zhang et al. [48] use a CycleGAN to translate labelled laboratory lubrication-state signals into field-style signals, sidestepping the need for labelled field-fault data entirely (Section 6.6.2). This addresses the lab-to-field distribution shift component of the data scarcity problem but, unlike SSL, still depends on a labelled laboratory source domain and has not been shown to generalise across bearing types or sensing modalities; SSL pre-training on unlabelled data and generative domain transfer from labelled laboratory data are therefore complementary rather than competing directions for data-scarce LCM.

Foundation model approaches – large pre-trained time-series models that can be adapted to new tasks with minimal retraining – represent a longer-horizon research direction. A bearing foundation model pre-trained on large-scale unlabelled vibration data from multiple sources and machine types could in principle be fine-tuned for LCM with very few labelled examples, substantially reducing the data collection burden. No such model exists for bearing monitoring as of the time of writing.

Future direction: SSL pre-training on large unlabelled AE or vibration datasets from bearings in normal operation – which are abundant in industrial monitoring archives – should be investigated as a method for learning representations informative for LCM. The key open questions are: (1) what pretext tasks are most effective for learning lubrication-relevant representations (time-frequency consistency, masked AE, contrastive clustering?); (2) how much labelled LCM data is required for effective fine-tuning; and (3) can SSL representations generalise across bearing types and operating conditions?

9 Conclusion

This review has provided a systematic examination of signal processing methods for lubrication condition monitoring of ball bearings, the first review in the LCM literature to organise its findings by signal processing method family rather than by sensing modality. A total of 48 primary sources were reviewed against inclusion criteria requiring online or in-line data acquisition, lubrication-specific monitoring objectives, and sufficient methodological detail to assess the signal processing approach.

On the sensing side, the review establishes a two-axis classification – active/passive and invasive/non-invasive – and shows that passive non-invasive methods (acoustic emission, vibration, passive ultrasound, microphone arrays, thermal monitoring) carry genuine lubrication-relevant information through the physical mechanisms summarised in Table 2. All of these signals increase in amplitude as the lubrication film thins and asperity contacts increase – a property that makes them inherently usable as lubrication condition indicators – but their amplitude is confounded by operating speed, load, and temperature independently of lubrication state. Active and invasive methods (ultrasonic reflectometry, electrical capacitance/impedance, optical, magnetic) provide more direct film thickness information via physics-based models, at the cost of installation complexity and, in some cases, bearing modification.

On the signal processing side, the master cross-reference matrix (Table 3) reveals a wide diversity of methods applied to LCM, spanning from computationally trivial time-domain statistics (RMS, pulse count rate) to sophisticated methods (graph neural networks, physics-informed κ regression). Time-domain statistical features dominate practical and industrial applications due to their low computational cost and robustness. Frequency-domain methods – particularly envelope analysis, spectral kurtosis, and fractional-octave filter banks – add mechanistic discrimination that time-domain methods cannot provide, enabling discrimination between starvation, contamination, and other lubrication fault types. Physics-based model-driven methods (spring-layer acoustic model, electrical circuit models, κ/Λ regression targets) bridge signal processing and lubrication physics, enabling quantitative film thickness estimation from sensor signals. Machine learning and data fusion methods show diagnostic promise but are at the research stage.

The comparative analysis in Section 7 reaches five principal conclusions: passive non-invasive methods are most sensitive to starvation, least sensitive to contamination, and confounded by operating condition variation; active methods provide quantitatively superior film thickness information; thermal monitoring responds more slowly than acoustic methods; no SP method has been validated under realistic variable-speed, variable-load industrial conditions; and the technology maturity gap between research and industry is large, with all sophisticated SP approaches remaining at the research or prototype stage.

Six signal-processing-centric research gaps are identified. The most consequential are the absence of standardised benchmark datasets that would enable rigorous cross-study comparison, the unsolved problem of speed- and load-invariant feature extraction for passive methods, and the underexploration of grease lubrication monitoring despite grease being the dominant lubricant in industrial rolling element bearings. Additional gaps include the disconnect between condition assessment outputs and actionable relubrication decisions, the lack of multi-sensor fusion validation in realistic industrial environments, and the untapped potential of self-supervised and foundation model approaches for learning informative representations from the large archives of unlabelled bearing monitoring data that exist in industrial settings.

The overarching conclusion is that the signal processing layer – not the sensing hardware – is the primary barrier limiting LCM from laboratory demonstration to industrial deployment. Advanced sensors capable of detecting lubrication-relevant signals exist and are demonstrated. What is lacking are SP methods that are simultaneously physically interpretable, robust to operating condition variation, validated under industrial conditions, and integrated with downstream maintenance decision systems.

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